

```
# éfjç» - ç” µè¬ï¼š188-5665-3017 - é,®ç®±ï¼šhs3434@foxmail.com -
ä, ºäº°ç½ ‘ç«™ï¼š[hs3434.github.io](https://hs3434.github.io) - **æ±
,è£Ææ,,âð “**ï¼šAMD ROCm Python âžÇç¬¹¼€ä ‘â • ¥ç¬¹ª, ^ï¼ˆROCm Radeon Cloud
æ¬¹âð ‘ï¼‰ --- ## ä, ºäº°ç®€ä» 3 â¹ ‘ç” Ýç®©äzjae¬¹â • ¥ç¬¹ªCE¬ç»éªCEï¼CE è ±ªà|è½¬âžç
K8s âðžÇç¬¹æ ^ã€, **â®¹â™¬ªCE¬éf¬¹²²ã€¬èµ,,æº¬èºfâº|ã€¬ä»»ãŠ;ç¼¬æž’ æ€GPU
æž¬çº + æœ¬ãŠ; **ã€, æœ€èèz ‘2 â ‘¬è ±ª| ROCm / K8s / Helm / FastAPI /
Prometheusi¼CEç¬¹ç«ªªCEæ^ [**k8s-llm-runtime**](https://github.com/hs3434/k8s-llm-
runtime)ï¼ˆK8s ä , Šçš,, vLLM æ¬¹jäžæœ¬ãŠ;ç¼½ ‘ä...³ï¼‰oi¼CE**v0.1.0
taggedï¼CEâ¬çžºªœº • ‘demo**ã€, ç†Ýæ,% Snakemake ç±» Argo Workflows çš,, ML
â¹³â¬ªæŠ½èè±ã€, --- ## æ±,è£ÆæŠ€èf½#### âðžÇç¬¹æ , àzf - Python 3.11+ - FastAPI - pytest
### â®¹â™¬¬ä , žç¼¬æž’ - Docker - Kubernetes - Helm - é • œáfä» “ªº “ç®jçº + ï¼ˆDaoCloud
é • œáfäšÉéYï¼% #### GPU / LLM ç” Ýæ€¬ - vLLM - AMD ROCm - PyTorch - Hugging Face ###
âð¬èš , æµªœ€š ä , žâ • ¥â... é “¾ - Prometheus - structlog - uv / ruff / mypy / GitHub Actions
- Linux èzç»´ --- ## é¹çºç»®éªCE ### k8s-llm-runtimeï¼šK8s ä , Šçš,, vLLM
æ¬¹jäžæœ¬ãŠ;ç¼½ ‘ä...³ï¼ˆv0.1.0 taggedï¼% **é—éç¬¹ï¼ˆª¬¹ªº” JD è£Æè´£ 1 +
3ï¼‰oi¼švLLM æ¬¹jäžéf¬¹²²éœ€ GPU + â®¹â™¬¬ + è • ¬ç” ±ï¼CEª¢é~Yâ¤šªºª...±ªº«æ—
¶ æ¬¹jäžçºjçº + âð¬æ¬,ï¼ˆç” çº ¶ /ª¬ª¢é~Yæf³ç” ” OpenAI â...¼â®¹ API æ— ¶ é—
¬æš »é«~ã€, ï¼ˆçº´ æž¥ª¬¹ªº” JDï¼šªªš æ¬¹jäžè®ç»f/æž¬çº +
Workshopã€¬æ • ™ç¬¹ªæ¬ðãŠYèf½æ¼” ç®oi¼% **æž ¶ æž,,**ï¼ˆªCE chart + ä , %ª±, lib +
é • œáfä¼¬ªCE¬ï¼‰oi¼š - `charts/llm-inference/`ï¼švLLM è ´Yè½² chartï¼CE**GPU vendor
â† ± æ¬ªä»...é¬ `values.yaml` ** ï¼ˆamd.com/gpu / nvidia.com/gpu / none
ä , %æ€¬ï¼CEvalues æ¬¾å¼¼¬è€%æ¬©ï¼% - `charts/llm-router/`ï¼šFastAPI ç¼½ ‘ä...³ chart -
ConfigMap æ³¬ª...¥ initContainer âš è½²½ª...,ª¼Y chart - RBAC æœ€ªº¬æ¬fé™¬ï¼ˆPod/
Service/ConfigMap read+write + Leaseï¼% - ServiceMonitor âð¬é€ªª¬ç” ”ï¼ˆService
0.0.0.0:9090/metricsï¼% - `src/k8s_llm_runtime/`ï¼š3 å±, lib æŠ½èè± -
`K8sJobOperator`ï¼ˆª½žª±,ï¼CEéšç” ” Job èºfâº|ï¼% - `VLLMIInferenceOperator`ï¼ˆª ,
ª±,ï¼CEhelm CLI subprocess + GPU vendorï¼% - `ModelOperator`ï¼ˆé«ª±,ï¼CEOpenAI â...
¼â®¹ + è ±ªš¬éf¬¹²²ï¼% **4 åªš æš€æœ¬ªºç,¹**ï¼š1. **lib ä , %ª±,ª^† å±,æŠ½èè±
j**ï¼ˆª¬¹ªº” JD âðžÇç¬¹æž ¶ æž,,èf½ãš;ï¼% - ä , šª±, ModelOperator âðªâ...³àzf
load_model(alias) / unload_model(alias) - ä ,ª±, VLLMIInferenceOperator âð¬è£... helm
ç» + èš , - à½žª±, K8sJobOperator éšç” ” K8s Job èºfâº| 2. **Helm ConfigMap æ³¬ª...¥
aliases**ï¼ˆª¬¹ªº” JD â®¹â™¬¬ªCE¬ + èµ,,æº¬èºfâº|ï¼% - éçzâ...¬æ¬æ¬ª¬æ¬¬ªžéf½
rebuild é • œáfä¼ˆGPU é • œáf 15-20GB rebuild 5-10mini¼% - ConfigMap
çš ’ çºšç” Ýæ • ^ï¼CEé ¶ é • œáfé ±ªªº 3. **K8s Lease â^† å , fâ¼¬é” ®**ï¼ˆª¬¹ªº”
JD é† ç¾èèçç»´ + HAï¼% - coordination.k8s.io/v1 Lease é²æç Router
â¤šª% æœ¬ª¹ ¶ âð¬ ‘éf¬¹²²ªCEä , €æ¬¹jäž¬ - é” ç²” âº|æCE% model
aliasï¼CE è ±ªš¬èèz æœY + ççº| 4. **ä , šãŠ;çºš Prometheus metrics**ï¼ˆª¬¹ªº” JD
â¹³â¬ªª¬¬èš , æµªœ€š ï¼% - `chat_completions_duration_seconds`ï¼ˆhistogramï¼CEª«
status code æ ±ç¾ai¼% - `http_requests_total`ï¼ˆcounterï¼CEæCE% method/route/statusï¼%
- `models_loaded`ï¼ˆgaugeï¼CEª¬æ¬ in-memory stateï¼% - ServiceMonitor âð¬é€ªºèz»
Grafana **GPU vendor â^ ± æ¬ª demo**ï¼ˆª¬¹ªº” JD âšª^† GPU é† ç¾èèï¼%oi¼š -
values.yaml `gpu.vendor=amd|nvidia|none` - AMDï¼šamd.com/gpu èµ,,æº¬èºfâº|ï¼CEROCm
vLLM é • œáf¬ - NVIDIAï¼šnvidia.com/gpu èµ,,æº¬èºfâº|ï¼CECUDA vLLM é • œáf¬ -
noneï¼šCPU only æ¬¹jä¼¬ï¼ˆkind cluster é»è®ï¼CEæœ¬ªœª¼€ä ‘ç” ”ï¼% **è
´¬è ±ªž¬èè¬**ï¼š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint
â...” clean - 2 Helm chartsï¼ˆllm-inference + llm-routerï¼% **è , ©âð¬ ‘ä , žžæ¬ªª**ï¼š -
Service DNS ä , ®ªCEé... bugï¼š --set fullnameOverride={release_name}` ä;ªª¬ - kind
```

eS,ç, 'a'½a + ...é • œâfâš ééYi¼šcontainerdConfigPatches + DaoCloud mirror -  
 èç™âºé½f½a + t™â...¥ ` docs/amd-interview-demo.md` ä¾¼çŽºâœºâ,è€f  
 \*\*æœªæ¥¥\*\*i¼v1.1i¼œi¼šSSE æµâ¼æž`ç + / RBAC æ” ¶ ç ´ s âªè » / çœYâž GPU  
 é + ç¾¼ e2e ### äYºäž Snakemake çš,, ML  
 â¹³âª • ¥ä½œæµæj + æž ¶ i¼^ç” Yä½èfæ™~i¼Eç±» Argo Workflowsi¼% -  
 èfæ™~i¼šSnakemake rule è`æ³ • æ— æ³ • âžžçŽºæ`jâ—ésæ,,ç»,â^ -  
 è¾¼èèj¼šæ ‘^â¼f rule è`æ³ • i¼Eç´ æž¥èfâ + ...éf` æž¥âEâš”æ€ç” Yæ^  
 rulei¼Eè s „èEè¾¼ “â...¥è¾¼ “â ± °ç±»âžžâª³â-ç s `æœ`â¼æ¹¼æž¥ - â...³è” i¼šJD  
 âš â^ + éj™æœªâ™~âjâ¹³âªç” â “ç»éâE”â€” â€” æœ-éj¹ç»æ~ ML â¹³âªæš½è±  
 jçš,,â • ¥ç`âžžè • µ ### è,, ‘æœªæž¥âE CNN è s Eç çâ™~i¼PyTorch i¼%â€” â€” 6 â “â -  
 |â¹ è • `çºçº s â ± ° - ç«`â^ç«`æ— ¶ çºâ • ç s `è s Eç EEG ä½jâ • - PyQt âžžæ— ¶ GUI +  
 FastAPI æš´éœ² - \*\*ä” “âº \*\*\*i¼š[github.com/hs3434/bci-pipeline-demo](https://github.com/  
 hs3434/bci-pipeline-demo) --- ## â • ¥ä½œç»âžž + ### ä, Šæµ • âº  
 ¢é ± Eâjâ½æ-ç s ‘æšEæœ%œ™â...-â, i½œ ç” Yç%œâ½jæ-â • ¥ç`â, ^ i½œ  
 2025.04 - è ± ³âš - \*\*Docker éf`ç½²\*\*i¼šä, šâšjâ»Eç Docker âE-i¼âYºäž snakemake +  
 è ± ç” æj + æž ¶ i¼% - \*\*æœªâšjç«`èçç»\*\*i¼šLinux æœªâšjâ™~æ—  
 ¥â, ç» æš i¼Enginx âââ “âEç + i¼Esystemd æœªâšjç¹ç +  
 \*\*âº ‘â¹³âª • ¥â... • â¼Eâ “\*\*i¼šä, šâšjè ± âš”âE-æµç»` + ç»æ³ • â • ¥â... • ###  
 ä, Šæµ • æ- s æ~ “ç” Yç%œâEâjç s ‘æšEæœ%œ™â...-â, i½œ  
 ç” Yç%œâ½jæ-ç” â “â • ¥ç`â, ^ i½œ 2022.08 - 2024.07 - \*\*âº ‘â¹³âªç” Yä½jâ • ¥â...  
 • â¼Eâ “\*\*i¼šR / Python / Linux / Docker / âº ‘âžžYç” Y - \*\*ä, šâšjè ± âš”âE-\*\*i¼šâEâ -  
 |è½-â½ • ç»,ä, šâšjæµç»`è ± âš”âE-i¼Eæ`jâ-âE-â€ââ-âµç»” -  
 \*\*â¹³âªœªâª\*\*i¼š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ###  
 ä, Šæµ • æ- s æ~ “ç” Yç%œâEâjæœ%œ™â...-â, i½œ ç” Yç%œâ½jæ-âžžâ¹ ç” Y i½œ  
 2021.10 - 2022.01 - â • ç» + èfžè½-â½ • ç»,i¼šè  
 ´ æž s æ€â, è s „â^ + æžæ€â” âžžæ€â, æ€ s âE-â^ + æž -  
 ä, »èæšEæœ~i¼šâ • ç» + èfžè½-â½ • ç»,æ€RæPythonæ€Linux --- ##  
 æ • ™è,²èfæ™~ \*\*è¥çâE-â + œæž-ç s ‘æšEâ s âj\*\* i½œ æµç»ç s ‘â -  
 |ä, ŽæšEæœ~ä, “ä, š i½œ æœ-ç s ‘ i½œ 2018.09 - 2022.07 i½œ GPAi¼š3.35 - æ` • ä, šèœæ-  
 ± i¼šâšEšâYºäžéœ»¾¼ ‘æ-è’ ,âžžâ½’ çjªšâªéªæ’ ä¼ç, â...³æ€ s çš ¶ âšæš—  
 â€” æ€ s çš ¶ æE ± æ ± æ€ - âµ s â^éj¹ç»i¼šæ¹ç»ææž¥ç s â¹ç`«èš±èœè “çæš—  
 é “œæ±jæY “çš,â½±â “â¼ââ,ä, Žâ½œèè...i¼% --- ## ä, âºâ¼~âšž - \*\*6 â “è ± âj K8s  
 âžžç«`æ ^\*\*i¼šâžžè ¶ â^ v0.1.0 tagged éj¹ç»i¼Eç-ç«`âEæ^âE Helm chart + Python  
 lib - \*\*çŽºâœº demo èf½âš, \*\*i¼šâ-è ± ââ, |ç” µè,, ‘æœ` kind cluster ä, Šè • ‘éšâEæ •  
 LLM æž`ç + æµç»` i¼é½²â + è • `ç” ± â + ` chat completions â + ` metricsi¼% -  
 \*\*âjâ¹ èf½âš,â¼\*\*i¼š3 â¹ â + ...âžžç” Yç%œâ½jæ-è • `ç • Eâ^ K8s  
 âžžç«`~i¼EæšEæœ~çfæf...é«` - \*\*3 â¹ Python â • ¥ç`âE-  
 ç»éâE\*\*i¼šäžžç s ‘ç” è,,šæœ-â^âº ‘â¹³âª • ¥â... • çš,âEæ • ´è • -â¾¼,,